

StadiumPulse Executive Pitch Sheet

1-MINUTE HACKATHON FINALS PRESENTATION GUIDE

THE PITCH FRAMEWORK (THREAT, GAP, NEED)

THE THREAT 🚨

Massive crowds create dangerous bottlenecks, security exposures, and logistical gridlocks during matches.

THE GAP ⚠️

Manual, isolated tools leave stadium command unable to adapt to sudden surges, blockages, or storm shifts.

THE NEED 🎯

Organizers urgently require an integrated command layer that unifies crowd flow and automates emergency response.

THE SOLUTION — STADIUMPULSE PLATFORM

Innovation Layer	How it Delivers Functional Fulfillment
Multi-Agent Agentic Loop	A Central Dispatcher routes queries to specialized Gemini sub-agents (Routing, Comms, Incident Commander) equipped with active Python tools to solve queue, transit, and weather logistics.
Live CV Analytics (Edge-decoupled)	Combines COCO-SSD body tracking and Face-API.js real-time facial expression classifiers inside the browser webcam stream, capturing real headcount and mood vibe percentages (Happy 😊 , Neutral 😐 , Frustrated 😡).
Live Digital Twin Map	Google Maps satellite view overlays live dynamic zone occupancy rings, wait times, and glowing warning alerts to give stadium operators high spatial awareness.
Automated Alert & PA Engine	Breach of occupancy thresholds triggers immediate incident alerts and automated PA scripts spoken live via synthetic speech audio broadcasts.

KEY DEMO WALKTHROUGH HIGHLIGHTS ("THE HAPPY PATH")

- Webcam Live Scanning:** Open the CV Demo. Start camera, smile, watch the Happy progress bar shoot to 95% and the Vibe Score rise. Sync the telemetry to a zone.
 - Immediate Dashboard Adaptation:** Confirm the stadium map ring scales and turns orange. Ask the chatbot: *"Where should I go to eat?"* The Routing Agent calls ``get_all_zones`` and recommends avoiding the busy hall.
 - Simulator Crisis Modeling:** Trigger a sudden downpour or exit blockage. Watch the dashboard glow red, generate critical PA announcers, and guide fans to safety exits.
- Build with AI Hackathon, Ahmedabad 2026
- StadiumPulse Pitch Sheet

Q&A & Technical Defense Guide

CRUSH THE JUDGES' TECHNICAL ASSESSMENT (15 POINTS)

STRATEGIC JUDGE Q&AS

Q1 How does your platform scale to handle 130,000 live spectators?

Response: We engineered a **decoupled Edge AI architecture**. High-bandwidth video frame processing is offloaded to client-side Edge GPUs via browser-compiled WebGL (using `TensorFlow.js` and `face-api.js`). **This results in 0 server compute cost and 0 cloud network latency**. The server only receives low-bandwidth headcount/emotion JSON vectors via a fast FastAPI backend, meaning a single lightweight container scales easily to thousands of concurrent users.

Q2 What happens if your Google Gemini API quota goes to zero during a crisis?

Response: Safety is our priority. We built a local **Smart Fallback Engine** (`_smart_fallback` in `agent.py`). If the Gemini API returns a 429 quota exception, our engine instantly intercepts the call, reads the active in-memory sensor database state, and synthesizes structured, accurate safety directions locally. **The system never goes dark, and safety guidance is always guaranteed.**

Q3 Why use a Multi-Agent architecture rather than one single prompt?

Response: Single-prompt architectures suffer from prompt bloat, high hallucination rates, and unstable tool-calling. By using a **high-speed Gemini 2.5 Flash Dispatcher**, we classify queries and delegate them to sub-agents with extremely narrow system instructions and custom tool subsets. This increases reasoning speed, ensures 100% stable tool calls, and allows us to use Gemini 2.5 Pro exclusively for the Incident Commander during high-stakes safety events, optimizing token costs.

Q4 How does the platform secure fan data and admin commands?

Response: **Privacy by design**: In our Edge AI pipeline, no video frames ever leave the browser—they are scanned locally, preserving fan anonymity. Admin simulator endpoints utilize strict Pydantic model validation. For production, these command endpoints would be bound to OAuth2 JWT tokens (RBAC) to ensure only authenticated incident commanders can trigger evacuation overrides.